

3(a)	Any 2 from: • particles / atoms / molecules / ions (very) close together / touching • regular, repeating pattern • vibrate about a fixed point	B2
3(b)	(much) greater <u>increase</u> in spacing of molecules (for vaporisation compared with fusion)	B1
3(c)(i)	−100 °C	B1

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Question	Answer	Marks
3(c)(ii)	time = $8.5 - 3.0$ = 5.5 min	C1
	$Pt = mL$ energy = power $\times$ time = $150 \times 5.5 \times 60$ = 49 500 J $L = \frac{E}{m}$ $= \frac{49500}{0.045}$	C1
	= 1100 kJ kg^{-1}	A1
3(c)(iii)	gas has a higher specific heat capacity (than liquid)	B1

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